PLANT SPECIES BIOLOGY

Plant Species Biology (2012) ••, ••–••

doi: 10.1111/j.1442-1984.2011.00346.x

Patchy distributions of *Spirogyra arcta* do not affect growth of the submerged macrophyte *Ceratophyllum demersum*

RUI-HUA LIU, BI-CHENG DONG, HONG-LI LI, QIAN ZHANG and FEI-HAI YU

College of Nature Conservation, Beijing Forestry University, No. 35 Qinghua East Road, Haidian District, Beijing 100083, China

Abstract

Spatial heterogeneity is common in aquatic conditions, but few studies have examined the effects of heterogeneous distributions of biological factors on aquatic plants. Spirogyra (*Spirogyra arcta*) coexists with many submerged macrophytes, such as *Ceratophyllum demersum*, but no study has examined the effects of heterogeneous distributions of spirogyra on the growth of submerged plants. We grew the submerged plant *C. demersum* in three homogeneous, aquatic conditions (0, 50 and 100% of the water surface in the container was evenly covered by spirogyra, referred to as 'control', '50%' and '100%', respectively) and two patchy conditions (the left half of the water surface in the container was not covered by spirogyra and 50% and 100% of the water surface in the right half of the container was evenly covered by spirogyra, referred to as '50%-patchy' and '100%-patchy', respectively). Compared with the control, the 100% treatment greatly decreased the biomass and number of ramets of *C. demersum*, but the 50% treatment did not. Growth of *C. demersum* in the left half of the container did not differ significantly between the control and the two heterogeneous treatments (50%-patchy and 100%-patchy). In addition, growth of *C. demersum* in the right half of the container did not differ between the 100% and the 100%-patchy treatment or between the 50% and the 50%-patchy treatment. Our results suggest that *C. demersum* can tolerate shading by spirogyra to a certain extent and that heterogeneous distributions of spirogyra do not affect its growth.

Keywords: aquatic plant, biological patchiness, coontail, spatial heterogeneity, spirogyra.

Received 25 February 2011; revision received 30 April 2011; accepted 19 May 2011

Introduction

Submerged macrophytes represent a special group of plants living in aquatic habitats (Cronk & Fennessy 2001; Vis *et al.* 2003; Chambers *et al.* 2008). Their growth and distribution are affected by many factors, including light (Van den Berg *et al.* 1998), water depth (Wallsten & Forsgren 1989), turbidity (Squires *et al.* 2002; Scheffer 2004), nutrients (Cronk & Fennessy 2001) and the distributions of other organisms (Strand & Weisner 1996; Weisner *et al.* 1997; Cronk & Fennessy 2001).

Spatial heterogeneity is common in nature (Alpert & Mooney 1996). In aquatic habitats, the existence of organisms such as phytoplankton (e.g. spirogyra), zooplankton, periphyton, emergent and/or floating plants can greatly reduce the amount of light reaching submerged plants and subsequently reduce their growth (Strand & Weisner 1996; Cronk & Fennessy 2001; Scheffer 2004). Moreover, the non-uniform distributions of these organisms can lead to a spatially heterogeneous distribution of light in the horizontal plane in water bodies (Scheffer 2004). To date, however, very few studies have examined how submerged macrophytes respond to spatial heterogeneity in resource supply (Idestam-Almquist & Kautsky 1995; Wolfer & Straile 2004; Xiao *et al.* 2006, 2007) and, to the best of our knowledge, no study has examined the effects of patchy distributions of biological factors such as phytoplankton, zooplankton, periphyton, emergent and/or floating plants on the growth of submerged plants.

Correspondence: Fei-Hai Yu
Email: feihaiyu@bjfu.edu.cn

© 2012 The Authors
Journal compilation © 2012 The Society for the Study of Species Biology

Clonal growth is common in aquatic plants (Les 1988; Grace 1993; Cronk & Fennessy 2001). The special traits (e.g. clonal integration) that clonal plants have enable them respond to environmental heterogeneity collectively (Alpert 1999; Saitoh *et al.* 2002; Yu *et al.* 2004, 2010; Chen *et al.* 2006; Roiloa *et al.* 2007; Zhang & He 2009a; b). These traits can help clonal plants effectively capture resources and thus increase the fitness of whole genets in heterogeneous environments (Birch & Hutchings 1994; Stuefer *et al.* 1994; Chen *et al.* 2010; Xiao *et al.* 2010). Few studies have examined the effects of environmental heterogeneity on the responses of submerged clonal plants. Because clonal plants dominate aquatic communities (Grace 1993; Klimes *et al.* 1997; Cronk & Fennessy 2001), understanding the phenotypic plasticity of aquatic clonal plants in response to heterogeneous aquatic conditions may give insights into the evolutionary significance of clonality in aquatic conditions.

To examine the effects of spatial heterogeneity of biological factors on the responses of submerged plants, we grew the submerged, clonal macrophyte *Ceratophyllum demersum* in three homogeneous aquatic conditions (with the whole water surface covered evenly by 0, 50% and 100% of *Spirogyra arcta*) and in two patchy aquatic conditions (with half of the water surface covered evenly by 50% or 100% of *S. arcta* and the other half not covered). Specifically, we addressed the following two questions: (i) under homogeneous conditions, does increasing cover of *S. arcta* decrease growth of *C. demersum*; and (ii) do heterogeneous distributions of *S. arcta* affect the performance of *C. demersum* plants?

Materials and methods

Study species

Ceratophyllum demersum L. (coontail) is a perennial, submerged aquatic plant belonging to the Ceratophyllaceae family, and it is distributed in various aquatic environments, including ponds, slow-flowing streams and ditches (Rodwell & Council 1998; DiTomaso & Healy 2003). *Ceratophyllum demersum* has firm, filament-like leaves (5–12 whorled at nodes). Mature plants of *C. demersum* rarely produce roots, but can anchor in sediment by buried stems (Gross *et al.* 2003; Lombardo & Cooke 2003). Its leaves and stems can absorb nutrients directly from the water (Gross *et al.* 2003; Lombardo & Cooke 2003). This species relies on vegetative propagation (e.g. stem fragments) or seeds to sustain populations in the summer, and produces turions in the winter (Eastman 1995; DiTomaso & Healy 2003). During the growth season, shoots of *C. demersum* can branch intensively and each shoot node (potential ramet) can develop into a new shoot (branch), so that a large clonal network can be formed. It can survive in low light conditions caused by, for example, green algae, most likely because it has a low light compensation point (Eastman 1995; Parsons & Cuthbertson 2001; Gross *et al.* 2003).

Spirogyra arcta Kuetz. (spirogyra) is a filamentous green algae belonging to the Zygnemataceae family that commonly grows in eutrophic water (John *et al.* 2002). Spirogyra can form dense filamentous masses via vegetative propagation, which greatly reduces light availability for submerged plants.

Plant material preparation

On 17 July 2010, *C. demersum* plants were collected from the lakes of the Winter Palace in Beijing and cultivated for 10 days in a greenhouse at Forestry Science, Beijing Forestry University. We selected mature shoots with an apex and at least 14 nodes. For each selected shoot, we cut off the additional nodes except the 12 closest to the apex and kept the 12-node shoot fragment in clear water. We then selected 60 shoot fragments of similar length; 40 of them were used for the experiment and the other 20 for initial measurements. The length of the fragments was 17 ± 1.7 cm (mean ± standard error [SE], $n = 20$) and the dry weight was 64.5 ± 0.2 mg.

On 24 July 2010, spirogyra was collected from an artificial pond in the Botanical Garden of the Institute of Botany, the Chinese Academy of Sciences, Beijing. It was cultivated in the greenhouse until the experiment began.

Experimental design

Forty plastic containers (40 cm long × 30 cm wide × 45 cm high) were used for the experiment; the walls of the containers were nearly non-transparent (reducing light by 99.1%; measured using a LI-250A light meter, Li-Cor, Lincoln, NE, USA). Each container was filled with 15 cm of soil (1:1 [v : v] mixture of sand and yellow loam and 0.1 g/L slow-release fertilizer [15N-11P-13K; Osmocote, the Scotts Company, Guangzhou Da Han Horticulture Development Company, Guangzhou, China]) as the sediment and 30 cm of tap water (Fig. 1). A plastic divider (4 cm high) was installed close to the water surface to a water depth of 2 cm (Fig. 1), dividing each container in half (i.e. the left half [L] and the right half [R]; Fig. 1). One shoot fragment of *C. demersum* was planted vertically in the center of each container with the two proximal nodes buried in the sediment and the shoot tip growing towards the divider (Fig. 1).

After 7 days of recovery, the *C. demersum* plants in the 40 containers were randomly assigned to the three homogeneous and two patchy treatments. In the first homogeneous treatment, the whole water surface was left uncovered (control); in the second homogeneous treatment, 50% of the water surface was evenly covered by

© 2012 The Authors
Journal compilation © 2012 The Society for the Study of Species Biology

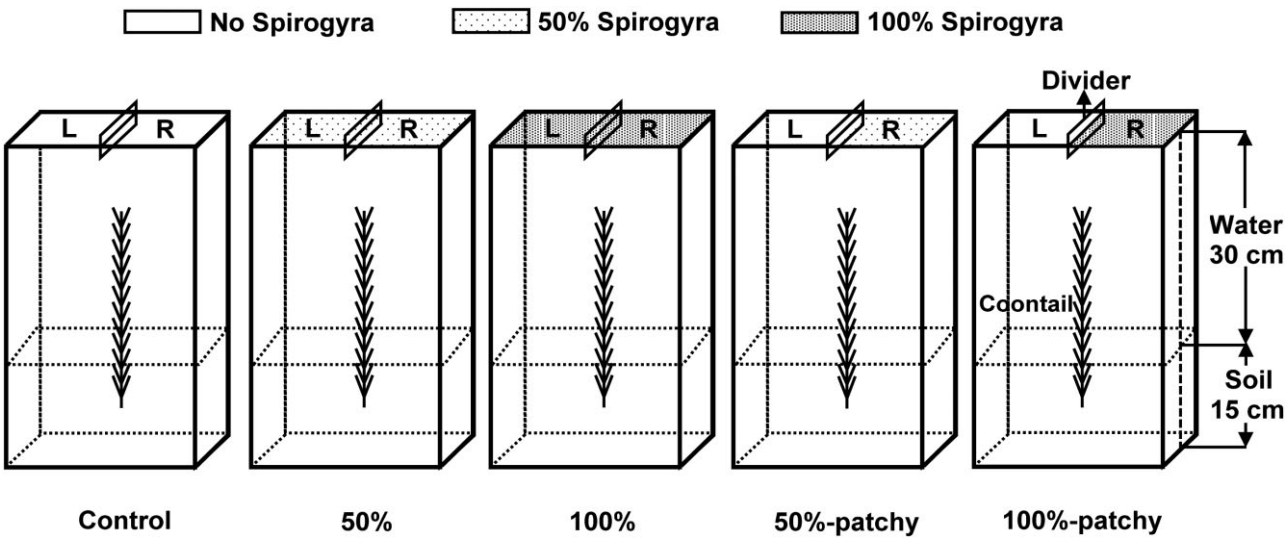

Fig. 1 Diagram showing the experimental design. A plastic divider (4 cm high) was installed close to the water surface to a water depth of 2 cm to divide each container into the left part (L) and the right part (R) and to restrict the movement of spirogyra (*Spirogyra arcta*) from one half to the other. One 12-node shoot fragment of coontail (*Ceratophyllum demersum*) was planted vertically in the center of each container with the two proximal nodes buried in the soil and the shoot tip growing towards the divider. In the three homogeneous treatments, the water surface in the container was not covered by spirogyra (control), or 50% or 100% of the water surface was evenly covered by spirogyra ('50%' and '100%', respectively); in the two heterogeneous treatments, the water surface on the left side of the container was not covered by spirogyra, but 50% or 100% of the water surface on the right side of the container was covered by spirogyra ('50%-patchy' and '100%-patchy', respectively).

S. arcta (50%); and in the third homogeneous treatment, 100% of the water surface was covered by *S. arcta* (100%; Fig. 1). In the patchy treatments, the water surface on the left half of the container was not covered by *S. arcta* and 50% or 100% of the water surface on the right half of the container was evenly covered by *S. arcta* (50%-patchy and 100%-patchy, respectively; Fig. 1). The divider could efficiently restrict movement of *S. arcta* from one part of the container to the other because *S. arcta* floats at the water surface, but it did not directly obstruct the growth and branching pattern of *C. demersum* (Fig. 1). There were eight replicates in each treatment.

The experiment was conducted in the greenhouse described above and lasted 10 weeks from 27 July to 5 October 2010. During the experiment tap water was supplied to maintain the same water depth in each container. The temperature and relative humidity in the greenhouse were 24.0 ± 0.4°C and 75.3 ± 1.3% (mean ± SE), respectively, measured hourly using two Hygrochron temperature loggers (iButton DS1923; Maxim Integrated Products, Dallas, TX, USA). During the experiment *C. demersum* did not flower and produced very few roots.

Measurements

At harvest, we separated the *C. demersum* plant in each container into three parts, the main shoot (the original shoot fragments plus elongation from the original shoot tip), the left part (shoot branches in the left part of the container) and the right part (shoot branches in the right part of the container). For each part, we measured the length and counted the ramets (nodes) of each shoot; total shoot length and number of ramets in one part were the sum of the corresponding variables of all shoots in that part. The plants in each part were then oven-dried at 70°C for 72 h and weighed. The biomass of the whole plant in each container was the sum of the biomass of the main shoot, the left part and the right part. Similarly, we calculated, for the whole plant, total stolon length and the number of ramets.

Data analysis

We used one-way ANOVAs followed by Dunnett's tests to compare differences in the number of ramets and shoot length of *C. demersum* among the five treatments. Before analysis, data on the number of ramets and shoot length were logarithmically transformed to increase the homogeneity of variance or normality. Data on biomass were analyzed using non-parametric tests. All analyses were conducted with SPSS 16.0 software (SPSS, Chicago, IL, USA).

Results

Biomass of the whole *C. demersum* plant (in the whole container) was smaller in the 100% treatment than in the

© 2012 The Authors
Journal compilation © 2012 The Society for the Study of Species Biology

other four treatments, but biomass in the control, 50%, 50%-patchy and 100%-patchy treatments did not differ significantly (Fig. 2a). The number of ramets of the whole plant in the 50% and 50%-patchy treatments was significantly larger than that in the 100% treatments, but the number in the control and 100%-patchy treatments did not differ significantly from that in the 50%, 100% or 50%-patchy treatments (Fig. 2b). Shoot length of the whole plant did not differ among the five treatments (Fig. 2c). None of the three growth measures of the whole plants differed significantly between the 50% and the 100%-patchy treatments (Fig. 2), suggesting that heterogeneous distributions of *S. arcta* did not affect growth of the whole *C. demersum* plants.

Biomass of the main shoot was significantly larger in the 50%-patchy treatment than in the control, but it did not differ significantly from that in the other three treatments (i.e. 50%, 100% or 100%-patchy; Fig. 3a). The number of ramets and shoot length of the main shoot did not differ among the five treatments (Fig. 3b,c).

Biomass of the left part of *C. demersum* (in the left half of the container) was significantly smaller in the 100% treatment than in the 50%, 50%-patchy and 100%-patchy treatments, but biomass in the control did not differ significantly from that in any of the other treatments (Fig. 4a). The number of ramets in the left part was largest in the 50% treatment, smallest in the 100% treatment and intermediate in the other three treatments (Fig. 4b). Shoot length of the left part did not differ significantly among the five treatments (Fig. 4c). None of the three measures of the left part differed significantly between the control and 50%-patchy treatment or between the control and 100%-patchy treatment (Fig. 4a–c), suggesting that heterogeneous distributions of *S. arcta* did not affect growth of the *C. demersum* plants in the left part.

Biomass of the right part of *C. demersum* (in the right half of the container) in the control was significantly larger than that in the 50%, 100% and 100%-patchy treatments, but it did not differ significantly from that in the 50%-patchy treatment (Fig. 4d). The number of ramets on the right part was largest in the control and the 50% treatment, smallest in the 100%-patchy treatment and intermediate in the 100% and 50%-patchy treatments (Fig. 4e). Shoot length of the right part did not differ significantly among the five treatments, but did show a similar trend (Fig. 4f). None of the three measures of the right part differed significantly between the 50% and 50%-patchy treatments or between the 100% and 100%-patchy treatments (Fig. 4d–f), suggesting that heterogeneous distributions of *S. arcta* did not affect growth of the *C. demersum* plants in the right part.

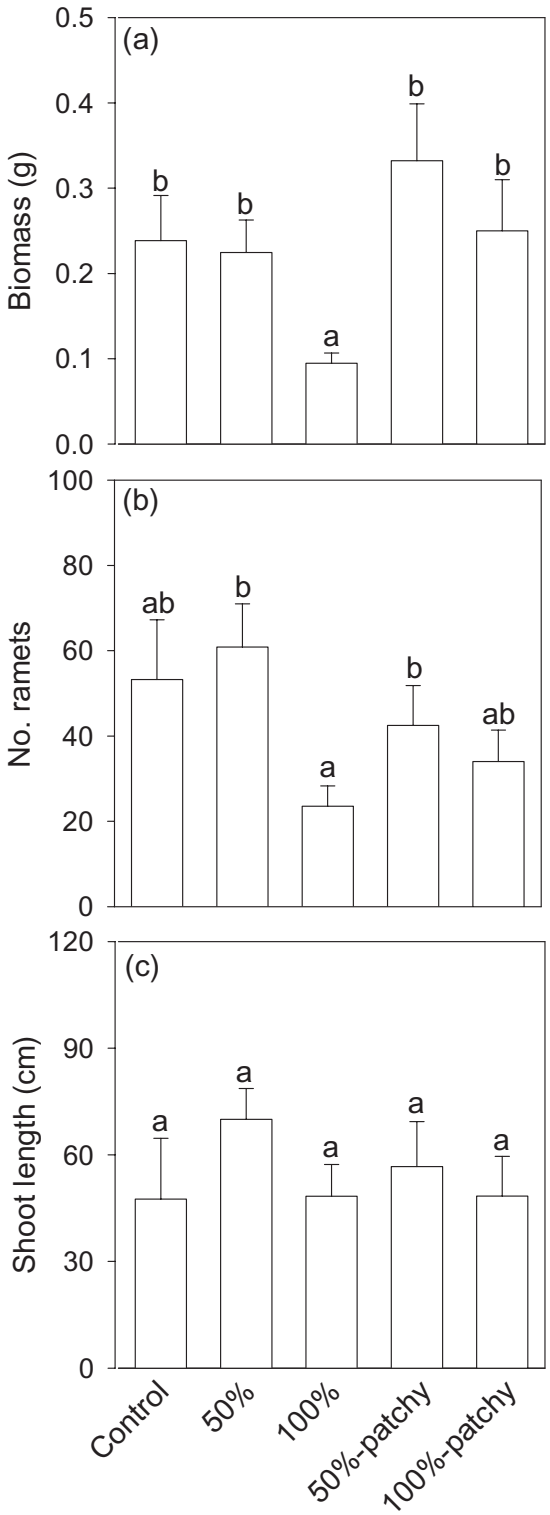

Fig. 2 Effects of homogeneous and heterogeneous distributions of spirogyra (*Spirogyra arcta*) on (a) biomass, (b) number of ramets and (c) shoot length of the whole plant of *Ceratophyllum demersum* (i.e. in the whole container). Bars sharing the same letters are not significantly different at $P = 0.05$. Values are mean + standard error.

© 2012 The Authors
Journal compilation © 2012 The Society for the Study of Species Biology

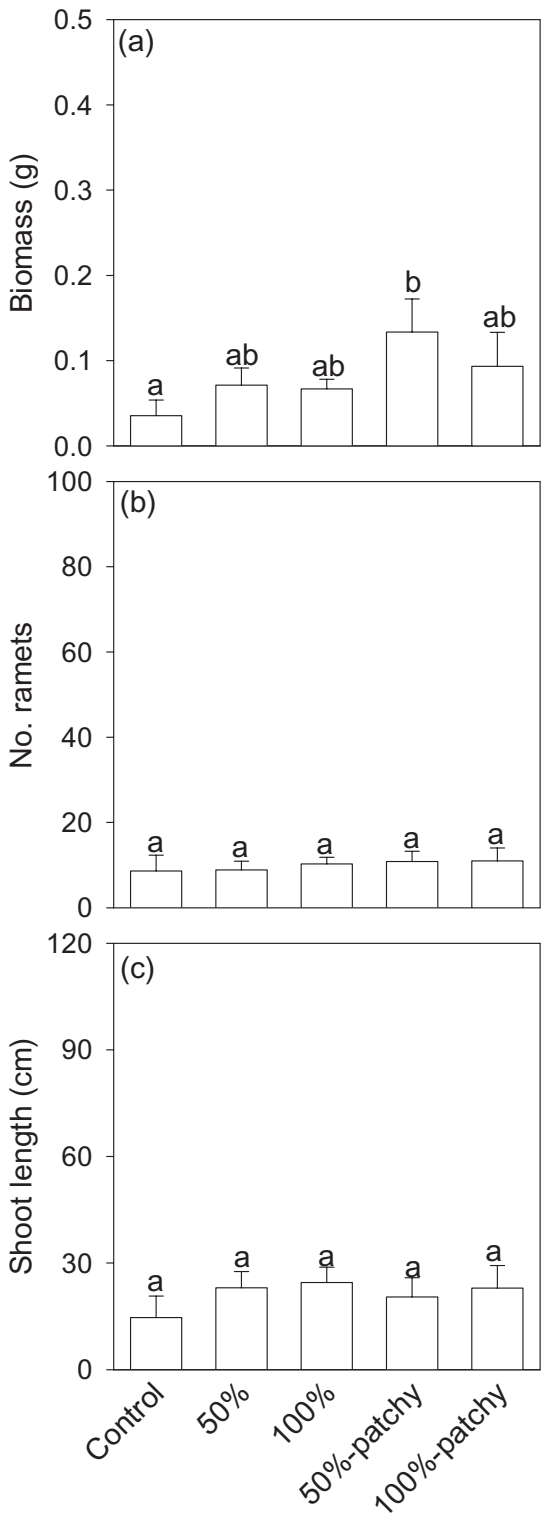

Fig. 3 Effects of homogeneous and heterogeneous distributions of spirogyra (*Spirogyra arcta*) on (a) biomass, (b) number of ramets and (c) shoot length of the whole plant of *Ceratophyllum demersum* (i.e. in the central part of the container). Bars sharing the same letters are not significantly different at $P = 0.05$. Values are mean + standard error.

Discussion

The results showed that when 100% of the water surface of the whole container was covered by floating spirogyra (*S. arcta*), the biomass and number of ramets of the submerged plant *C. demersum* were greatly suppressed. This is likely because the light that reached the *C. demersum* plants was greatly reduced because of shading by spirogyra. The reduced light restrained the photosynthetic capacity of *C. demersum* and thus reduced growth. Another possible reason was that spirogyra used a large amount of nutrients (Lei *et al.* 2007; Ma & Lei 2008) so that the availability of nutrients for *C. demersum* was low and thus growth was reduced (Spencer & Wetzel 1993; Lombardo & Cooke 2003). However, all *C. demersum* plants survived to the end of the experiment even when the whole water surface was covered by spirogyra, and compared with the control (no spirogyra), 50% water surface coverage by spirogyra did not affect the biomass or number of ramets of *C. demersum*. Thus, *C. demersum* can exhibit a strong tolerance to the relatively low light conditions caused by spirogyra, most likely because its light compensation point is low (Spencer & Wetzel 1993). Because the experiment was relatively short (only 10 weeks), it is possible that all *C. demersum* plants in the treatment in which spirogyra covers the whole area of the water surface will die if the experiment continued much longer.

If the spatial distribution of spirogyra had significant effects on *C. demersum*, we would find that the growth of the left part of *C. demersum* differed significantly between the control and the two heterogeneous treatments (50%-patchy and 100% patchy), or that the growth of the right part differed significantly between the 100% and the 100%-patchy treatment or between the 50% and the 50%-patchy treatment, or that the growth of the whole plant differed between the 50% and the 100%-patchy treatment because the total cover of spirogyra was the same in these two treatments. However, none of the three growth measures differed significantly between these treatments, suggesting that heterogeneous distributions of spirogyra did not affect the growth of *C. demersum*. In contrast, many studies in terrestrial environments have shown that spatial heterogeneity greatly affects the growth of plants (Alpert 1999; van Kleunen & Stuefer 1999; Alpert *et al.* 2003; Yu *et al.* 2004, 2010; Wang *et al.* 2008, 2009), and ramets growing in shade could benefit from the interconnection to ramets growing in full light (Hartnett & Bazzaz 1983; Slade & Hutchings 1987; Stuefer *et al.* 1994; Dong 1995; Wijesinghe & Hutchings 1996; Alpert 1999; Saitoh *et al.* 2002; Alpert *et al.* 2003). Previous studies have found that spatial heterogeneity in light and/or nutrient supply can greatly affect the growth of two submerged macrophytes, *Vallisneria spiralis* (Xiao *et al.* 2006, 2007) and

© 2012 The Authors
Journal compilation © 2012 The Society for the Study of Species Biology

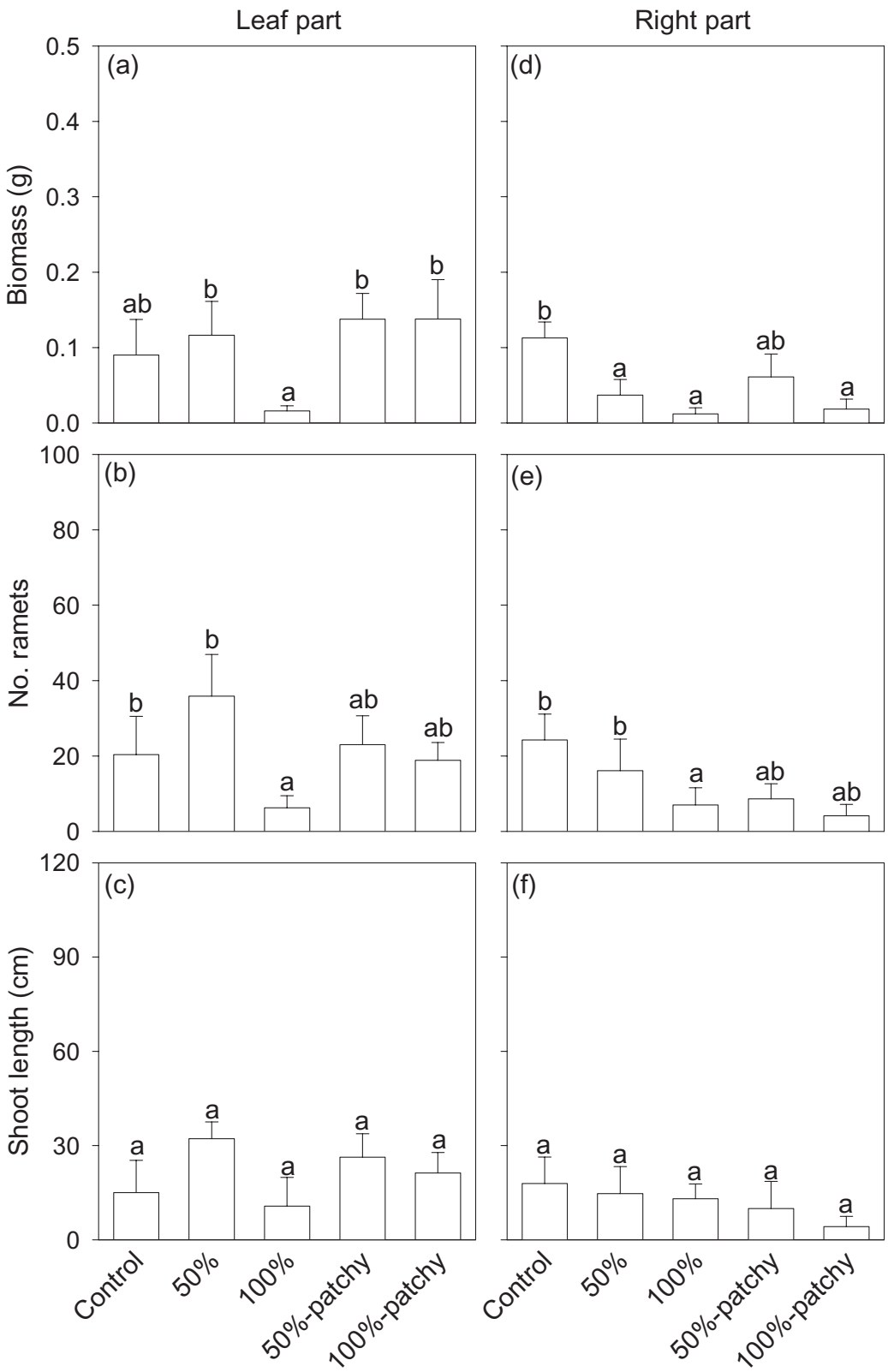

Fig. 4 Effects of homogeneous and heterogeneous distributions of spirogyra (*Spirogyra arcta*) on (a,d) biomass, (b,e) number of ramets and (c,f) shoot length of the left part (a–c) and the right part (d–f) of *Ceratophyllum demersum*. Bars sharing the same letters are not significantly different at $P = 0.05$. Values are mean + standard error.

© 2012 The Authors
Journal compilation © 2012 The Society for the Study of Species Biology

Potamogeton perfoliatus (Idestam-Almquist & Kautsky 1995; Wolfer & Straile 2004).

In the present study, the irresponsiveness of *C. demersum* to the heterogeneous distributions of spirogyra is likely because the effects of spirogyra were collective and the existence of spirogyra changed not only light penetration, but also many other environmental factors (e.g. nutrients, turbidity, dissolved oxygen and temperature) in the aquatic habitat (Sand-Jensen & Sondergaard 1981; Sand-Jensen 1990; Spencer & Wetzel 1993; Hu *et al.* 2000; Lombardo & Cooke 2003). Spirogyra can absorb small particles and thus reduce the turbidity of water, increasing the light that reaches the submerged plants under the water. This effect may greatly counteract the negative effects of shading by spirogyra on the submerged plant, and as a result we cannot find a significant effect of spatial heterogeneity on the distribution of spirogyra on *C. demersum*. Another explanation for the irresponsiveness to the heterogeneous distributions of spirogyra may be a species-specific effect of *C. demersum*. It is possible that *C. demersum* evolved a relatively consolidate strategy and thus does not respond to environmental heterogeneity.

This study is the first to test the effects of spatial heterogeneity in the distribution of spirogyra on the growth of submerged plants. Based on the results of this study, there is no indication that patchy distributions of spirogyra affect the growth of *C. demersum*. However, because the existence of spirogyra can change many environmental factors in aquatic ecosystems (Sand-Jensen & Sondergaard 1981; Sand-Jensen 1990; Spencer & Wetzel 1993; Hu *et al.* 2000; Lombardo & Cooke 2003), further studies should separate the effects of shading by spirogyra from other effects, such as changing water turbidity or reducing nutrients and water temperature by spirogyra, to fully understand the interaction between spirogyra and submerged plants.

Acknowledgments

We thank Chun-Fu Zhao for assistance with experiment preparation, Ying Zhou and Chun Zhu for help at harvest, and Professor Wei Li for valuable advice with regard to the experimental setup. We also thank the handling editor for recommending the use of a Dunnett's test for multiple comparisons. This research is supported by the Fundamental Research Funds for the Central Universities (JC2011-4), the Forestry Commonwealth Project (201004078), The National Natural Science Foundation of China (31070371) and the Forestry Soft-Science Research Project (2008-R02).

References

Alpert P. (1999) Clonal integration in *Fragaria chiloensis* differs between populations: ramets from grassland are selfish. *Oecologia* **120**: 69–76.

Alpert P., Holzapfel C. & Slominski C. (2003) Differences in performance between genotypes of *Fragaria chiloensis* with different degrees of resource sharing. *Journal of Ecology* **91**: 27–35.

Alpert P. & Mooney H. A. (1996) Resource heterogeneity generated by shrubs and topography on coastal sand dunes. *Vegetatio* **122**: 83–93.

Birch C. P. D. & Hutchings M. J. (1994) Exploitation of patchily distributed soil resources by the clonal herb *Glechoma hederacea*. *Journal of Ecology* **82**: 653–664.

Chambers P., Lacoul P., Murphy K. & Thomaz S. (2008) Global diversity of aquatic macrophytes in freshwater. *Hydrobiologia* **595**: 9–26.

Chen J.-S., Lei N.-F. & Dong M. (2010) Clonal integration improves the tolerance of *Carex praeclara* to sand burial by compensatory response. *Acta Oecologica* **36**: 23–28.

Chen J.-S., Lei N.-F., Yu D. & Dong M. (2006) Differential effects of clonal integration on performance in the stoloniferous herb *Duchesnea indica*, as growing at two sites with different altitude. *Plant Ecology* **183**: 147–156.

Cronk J. K. & Fennessy M. S. (2001) *Wetland Plants: Biology and Ecology*. Lewis Publishers, Boca Raton.

DiTomaso J. M. & Healy E. A. (2003) *Aquatic and Riparian Weeds of the West*. University of California, Division of Agriculture and Natural Resources, California.

Dong M. (1995) Morphological responses to local light conditions in clonal herbs from contrasting habitats, and their modification due to physiological integration. *Oecologia* **101**: 282–288.

Eastman J. (1995) *The Book of Swamp and Bog: Trees, Shrubs, and Wildflowers of the Eastern Freshwater Wetlands*. Stackpole Books, Mechanicsburg.

Grace J. B. (1993) The adaptive significance of clonal reproduction in angiosperms: an aquatic perspective. *Aquatic Botany* **44**: 159–180.

Gross E. M., Erhard D. & Ivanyi E. (2003) Allelopathic activity of *Ceratophyllum demersum* L. and *Najas marina* ssp. *intermedia* (Wolfgang) Casper. *Hydrobiologia* **506**: 583–589.

Hartnett D. C. & Bazzaz F. A. (1983) Physiological integration among intraclonal ramets in *Solidago canadensis*. *Ecology* **64**: 779–788.

Hu J.-Z., Sun X.-W., Lv Y.-Z. & Yang W.-Y. (2000) Shui dao shui mian fa sheng, xiao zhang ji fang zhi ji shu. *Reclaim and Rice Cultivate* **3**: 3–8.

Idestam-Almquist J. & Kautsky L. (1995) Plastic responses in morphology of *Potamogeton pectinatus* L. to sediment and above-sediment conditions at two sites in the northern Baltic proper. *Aquatic Botany* **52**: 205–216.

John D., Whitton B., Brook A. (2002) *The Freshwater Algal Flora of the British Isles: An Identification Guide to Freshwater and Terrestrial Algae*. Cambridge University Press, Cambridge.

Klimes L., Klimesova L., Hendriks R. & van Groenendael J. (1997) Clonal plant architecture: a comparative analysis of form and function. In: de Kroon H. & van Groenendael J. (eds). *The Ecology and Evolution of Clonal Plants*. Backhuys, Leiden, pp. 1–29.

Lei G., Fan W., Li Y., Deng H. & Li J. (2007) The simulated research on removing nitrogen and phosphorous by immobilized algal film. *Ecology and Environment* **16**: 1166–1170.

Les D. H. (1988) Breeding systems, population structure, and evolution in hydrophilous angiosperms. *Annals of the Missouri Botanical Garden* **75**: 819–835.

© 2012 The Authors
Journal compilation © 2012 The Society for the Study of Species Biology

Lombardo P. & Cooke G. D. (2003) *Ceratophyllum demersum*–phosphorus interactions in nutrient enriched aquaria. *Hydrobiologia* **497**: 79–90.

Ma J. & Lei G.-Y. (2008) Characteristics of phosphorus removal and growth inhibition of micro-algal species by *Spirogyra*. *Acta Scientiae Circumstantiae* **28**: 476–483.

Parsons W. T. & Cuthbertson E. G. (2001) *Noxious Weeds of Australia*. CSIRO Publishing, Collingwood.

Rodwell J. & Council N. C. (1998) *British Plant Communities: Aquatic Communities, Swamps and Tall-Herb Fens*. Cambridge University Press, Cambridge.

Roiloa S. R., Alpert P., Tharayil N., Hancock G. & Bhowmik P. C. (2007) Greater capacity for division of labour in clones of *Fragaria chiloensis* from patchier habitats. *Journal of Ecology* **95**: 397–405.

Saitoh T., Seiwa K. & Nishiwaki A. (2002) Importance of physiological integration of dwarf bamboo to persistence in forest understorey: a field experiment. *Journal of Ecology* **90**: 78–85.

Sand-Jensen K. (1990) Epiphyte shading: its role in resulting depth distribution of submerged aquatic macrophytes. *Folia Geobotanica* **25**: 315–320.

Sand-Jensen K. & Sondergaard M. (1981) Phytoplankton and epiphyte development and their shading effect on submerged macrophytes in lakes of different nutrient status. *Internationale Revue Der Gesamten Hydrobiologie Und Hydrographie* **66**: 529–552.

Scheffer M. (2004) *Ecology of Shallow Lakes*. Kluwer Academic Publishers, Dordrecht.

Slade A. J. & Hutchings M. J. (1987) The effects of light intensity on foraging in the clonal herb *Glechoma hederacea*. *Journal of Ecology* **75**: 639–650.

Spencer W. E. & Wetzel R. G. (1993) Acclimation of photosynthesis and dark respiration of a submersed angiosperm beneath ice in a temperate lake. *Plant Physiology* **101**: 985–991.

Squires M., Lesack L. & Huebert D. (2002) The influence of water transparency on the distribution and abundance of macrophytes among lakes of the Mackenzie Delta, Western Canadian Arctic. *Freshwater Biology* **47**: 2123–2135.

Strand J. A. & Weisner S. E. B. (1996) Wave exposure related growth of epiphyton: implications for the distribution of submerged macrophytes in eutrophic lakes. *Hydrobiologia* **325**: 113–119.

Stuefer J. F., During H. J. & de Kroon H. (1994) High benefits of clonal integration in two stoloniferous species, in response to heterogeneous light environments. *Journal of Ecology* **82**: 511–518.

Van den Berg M. S., Coops H., Simons J. & de Keizer A. (1998) Competition between *Chara aspera* and *Potamogeton pectinatus* as a function of temperature and light. *Aquatic Botany* **60**: 241–250.

van Kleunen M. & Stuefer J. F. (1999) Quantifying the effects of reciprocal assimilate and water translocation in a clonal plant by the use of steam-girdling. *Oikos* **85**: 135–145.

Vis C., Hudon C. & Carignan R. (2003) An evaluation of approaches used to determine the distribution and biomass of emergent and submerged aquatic macrophytes over large spatial scales. *Aquatic Botany* **77**: 187–201.

Wallsten M. & Forsgren P. (1989) The effects of increased water level on aquatic macrophytes. *Journal of Aquatic Plant Management* **27**: 32–37.

Wang N., Yu F.-H., Li P.-X., He W.-M., Liu F.-H., Liu J.-M. & Dong M. (2008) Clonal integration affects growth, photosynthetic efficiency and biomass allocation, but not the competitive ability, of the alien invasive *Alternanthera philoxeroides* under severe stress. *Annals of Botany* **101**: 671–678.

Wang N., Yu F.-H., Li P.-X., He W.-M., Liu J., Yu G.-L., Song Y.-B. & Dong M. (2009) Clonal integration supports the expansion from terrestrial to aquatic environments of the amphibious stoloniferous herb *Alternanthera philoxeroides*. *Plant Biology* **11**: 483–489.

Weisner S. E. B., Strand J. A. & Sandsten H. (1997) Mechanisms regulating abundance of submerged vegetation in shallow eutrophic lakes. *Oecologia* **109**: 592–599.

Wijesinghe D. K. & Hutchings M. J. (1996) Consequences of patchy distribution of light for the growth of the clonal herb *Glechoma hederacea*. *Oikos* **77**: 137–145.

Wolfer S. R. & Straile D. (2004) Spatio-temporal dynamics and plasticity of clonal architecture in *Potamogeton perfoliatus*. *Aquatic Botany* **78**: 307–318.

Xiao Y., Tang J., Qing H., Ouyang Y., Zhao Y., Zhou C. & An S. (2010) Clonal integration enhances flood tolerance of *Spartina alterniflora* daughter ramets. *Aquatic Botany* **92**: 9–13.

Xiao K., Yu D. & Wang J. (2006) Habitat selection in spatially heterogeneous environments: a test of foraging behaviour in the clonal submerged macrophyte *Vallisneria spiralis*. *Freshwater Biology* **51**: 1552–1559.

Xiao K., Yu D., Xu X. & Xiong W. (2007) Benefits of clonal integration between interconnected ramets of *Vallisneria spiralis* in heterogeneous light environments. *Aquatic Botany* **86**: 76–82.

Yu F.-H., Dong M. & Krusi B. (2004) Clonal integration helps *Psammochloa villosa* survive sand burial in an inland dune. *New Phytologist* **162**: 697–704.

Yu F.-H., Wang N., He W.-M. & Dong M. (2010) Effects of clonal integration on species composition and biomass of sand dune communities. *Journal of Arid Environments* **74**: 632–637.

Zhang L.-L. & He W.-M. (2009a) Consequences of ramets helping ramets: no damage and increased nutrient use efficiency in nurse ramets of *Glechoma longituba*. *Flora* **204**: 182–188.

Zhang L.-L. & He W.-M. (2009b) Spatial covariance in resources affects photosynthetic rate and water potential, but not the growth of *Glechoma longituba* fragments. *Flora* **204**: 628–634.

© 2012 The Authors
Journal compilation © 2012 The Society for the Study of Species Biology